DO NOW — RECALL

Lesson 061 · Thinking Ahead: Inputs, Preconditions and

Name: _____ Class: _____ Date: _____

1. What is an abstraction assumption?

2. Why validate inputs before processing?

3. What is a reusable component?

4. Array a starts as [2,3,5,7]. Execute a[1]=a[0]+a[2], then a[2]=a[1]-a[3]. Trace the final array, using zero-based indexing.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A simplifying rule in a model.

2. Model answer

Invalid values can violate preconditions.

3. Model answer

A method/module designed for use in several parts of software.

4. Model answer

[2,7,0,7]. The first assignment sets a[1] to 7; the second uses that updated value, so a[2]=7-7=0.

By the end, you can…

• Specify the inputs and outputs before choosing algorithms: ticket count and unit price determine total charge.

• Make preconditions explicit. Invalid negatives or overflow can produce incorrect outputs; validation and appropriate types reduce risks.

• Caching saves a previously computed result for reuse, trading memory and stale-result risk against repeated calculation time. Reusable methods reduce repeated code.

Vocabulary — know these terms

precondition — Requirement for valid use.

postcondition — Expected condition after execution.

cache — Saved result for quicker reuse.

module — Reusable program component.

NEW LEARNING

Read, annotate and explain

Identify

Specify the inputs and outputs before choosing algorithms: ticket count and unit price determine total charge.

Model

Make preconditions explicit. Invalid negatives or overflow can produce incorrect outputs; validation and appropriate types reduce risks.

Evaluate

Caching saves a previously computed result for reuse, trading memory and stale-result risk against repeated calculation time. Reusable methods reduce repeated code.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A function doubles an integer. State input/output and a precondition if it accepts values 0–50 only.

Explain the decision

The algorithm's result can be correct mathematically but invalid under the declared application rules.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For function TicketCost(count,price), state two inputs, output and one precondition. [5]

Question 2

Give a scenario where caching a result saves time, plus a stale-data risk and one invalidation strategy. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

An API-dependent school timetable app calls same room lookup 200 times. Design a reusable method with cache and describe two tests. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Caching a result can never affect correctness.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.